BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LARKANA

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2017

PHYSICS

Past Paper 2017

Retyped faithfully from the original board paper — same questions, no changes.

Part-A Multiple Choice Questions (MCQs)(Type A)

Time Allowed: 20 Minutes | **Maximum Marks:** 16

Note: Identify the correct answer of each MCQ from its given four options and write on main Answer script/copy.

Q1. The vernier calipers is used to measure distance up to..

(A) 0.5 mm
(B) 0.005 mm
(C) 0.0005 mm
(D) None

Q2. In S.I units, the Newton-metre is unit of...

(A) Velocity
(B) Force
(C) Acceleration
(D) None

Q3. Fc = ...

(A) mv^2/r
(B) $m4\pi^2r/T^2$
(C) Body
(D) None

Q4. A ..... = nf

(A) v
(B) in
(C) I2
(D) None

Q5. Buoyant force is equal to the.....of liquid displaced by the body.

(A) Volume
(B) Weight
(C) Pressure
(D) Density

Q6. If p=4 dioptre then f=.....

(A) 20 cm
(B) 25 cm
(C) 10 cm
(D) None

Q7. The ratio between output and input is called.....of machine.

(A) Mechanical advantages
(B) Efficiency
(C) Stress
(D) Strain

Q8. Unit of co efficient of viscosity is.....

(A) Centipoises
(B) Density
(C) Cusec
(D) None

Q9. The ratio between sin of Li and Sin of Lr is called.....

(A) Snell's Law
(B) Coulomb's Law
(C) Ohm's Law
(D) Boyle's Law

Q10. [illegible in source scan] = I1 - I2

(A) I3
(B) I1
(C) I2
(D) None

Q11. In equation F=Kx, K is called spring constant having unit.....

(A) Nm^{-1}
(B) F/L
(C) [illegible in source scan]
(D) m/s

Q12. Work will be zero if force acts at.....angle.

(A) Straight angle
(B) Right angle
(C) Acute angle
(D) Obtuse angle

Q13. Gravitational force of attraction between body and earth is equal to.....of body.

(A) Weight
(B) Mass
(C) Volume
(D) Density

Q14. Law of heat exchange was first enunciated by.....

(A) Prevost
(B) Pascal
(C) Newton
(D) None

Q15. The loss of a large amount of heat energy, causes only small drop in the ..... of water.

(A) Temperature
(B) Mass
(C) Both
(D) None

Q16. The potential difference between two point is 110V and work done by charge 550J then charge(q) will be.....

(A) 6C
(B) 7C
(C) 8C
(D) 5C

Part-B Constructed Response Questions (CRQs)

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 45

> Note: Attempt any Nine(09) questions from Section-B, And Two(02) questions from Section-C.

Q1. Explain force of action and reaction for a person sitting on chair.

Q2. Speed of light in diamond is 1.24×10^5 Km/s and that in air 3.0×10^8 m/s. Calculate the refractive index of diamond.

Q3. Describe A.C and D.C with the help of diagram. How direct current is changed to alternate current?

Q4. Can a force directed to north balance a force directed to east? Explain.

Q5. Calculate the quantity of current passing through cross-sectional area 0.8C in 10 seconds.

Q6. Is it possible for a body to be accelerated if its speed is constant?

Q7. Describe the similarities between Magnetism and Electricity.

Q8. Does a hydrogen filled balloon possesses any potential energy explain

Q9. Why are explosives sound produced in the sun not heard on the earth

Q10. A metal bar of cross section area $1000mm^2$ is 3 m long. If a force increases its length by 0.24mm find the force. If young's modulus of the metal is 8.0×10^{10} N/m^2.

Q11. Describe the Law of Momentum and its anyone application.

Q12. Define Shunt. How Galvano meter is converted into Ammeter?

Part-C (Extended Response Questions (ERQs))(Long Answer Questions)

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 24

Q13. Describe Irregular Reflection of Light and its importance in detail.

Q14. (a) Describe Circuit conventional current, Actual flow of current and its unit. (b) Prove that $s = v \times 2\pi r$.

Q15. Describe Echo at what distance between person and the obstacle it is heard? Describe the application of Echo used by the Doctors.